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Yu Saito (齋藤 佑, he/his/him)

Curriculum Vitæ

Visiting Researcher (Kawai Group) · **Kanagawa University**
Ph.D. in Eng. | Magnetic Resonance / Quantum Sensing

Academic Profile

I am a Visiting Researcher at the Research Institute for Integrated Science, Kanagawa University (Kawai Group). After earning my Ph.D. in Engineering from Yamagata University, I have consistently dedicated my research career to the field of ***Magnetic Resonance, with a core expertise in various Electron Spin Resonance (ESR) techniques***. My work spans a wide range of disciplines—addressing both fundamental and applied challenges. Drawing on this extensive background, I am currently focused on ***pioneering new frontiers by integrating quantum phenomena with Machine Learning (ML)*** to achieve advanced sensing and measurement capabilities.

If you have any questions, please feel free to contact me.

Keywords: ESR/EPR, Quantum Sensing/Information, Python, ODMR

Contact

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Location:
Kanagawa University, Yokohama, Japan

Online Profiles

Google Scholar
ORCID
ResearchGate
LinkedIn
GitHub
Twitter (X)
e-Rad: 60772142

Education

2012 – 2015 **Doctor of Philosophy in Engineering** | Bioengineering, Yamagata University, Japan

Doctoral Dissertation: Studies on Individual Reactive Oxygen Species Generation Systems and Their Application to the Evaluation of Reactive Oxygen Scavenging Activity

Advisor: Prof. Tateaki Ogata, Ph.D.

2008 – 2012 **Master of Engineering** | Chemistry and Chemical Engineering, Yamagata University, Japan

Thesis: Study on Singlet Oxygen Generation Systems in Non-aqueous Media and Its Application to Antioxidant Capacity Evaluation

Advisor: Prof. Tateaki Ogata, Ph.D.

2004 – 2008 **Bachelor of Engineering** | Materials Science and Engineering, Akita University, Japan

JABEE accredited program completed

Thesis: Changes in Optical Properties of Retinoic Acid LB Films on ITO Films and Physical Properties of ITO Films Prepared by Sol-Gel/Dip-Coating Method

Advisor: Lect. Yutaka Tsujiuchi, Ph.D.

2002	Voluntary withdrawal | Electronic Engineering, Tohoku Institute of Technology, Japan

Work Experience & Academic Positions

November 2025	**Specially Appointed Postdoctoral Researcher** Moonshot Project (Shu Lab.), International Christian University (ICU)
August 2024	**Special-appointed Postdoctoral Researcher**, Tokyo Institute of Technology (Institute of Science Tokyo)
July 2024 **Present** **April 2023**	**Visiting Researcher**, Research Institute for Integrated Science (Kawai Group), Kanagawa University
September 2023	**PJPD Postdoctoral Fellow**, Special-appointed NEDO Project Postdoctoral Fellow, Sophia University
July 2023 **June 2021**	**Research Support Specialist**, Cryogenic Division, Research Facility Center for Science and Technology (March 2017 – September 2018, April 2019 – May 2021) **Lecturer (Research Fellow)**, Molecular Photoscience Research Center (October 2018 – March 2019) Kobe University
May 2021 **March 2017**	**Postdoctoral Researcher**, Yamagata University (Biochemical Engineering)
February 2015 **May 2012**	**Research Assistant**, Yamagata University
February 2013 **May 2010**	**Project Science Instructional and Laboratory Assistant** Japan Science and Technology Agency (JST) Science Education Support Staff Assignment Program, Yonezawa City Board of Education

Awards & Fellowship

2015	**JASSO Recognition for Research Excellence (Half Loan Repayment Exemption)** Selected as a top-tier student for outstanding research achievements and academic excellence. Japan Student Services Organization (JASSO)
September 2013	**Excellent Poster Award**, International Symposium for the 70th Anniversary of the Tohoku Branch of the Chemical Society of Japan The Tohoku Branch of the Chemical Society of Japan

Projects Grants Funding

2021	**Joint Research of Research Center for Development of Far-Infrared Region, University of Fukui (FIR-UF)** Project Number: R03FIRDG029A Project Name: ESR Study of Cerium Magnesium Nitrate under Extreme Low Temperature. Grant Amount: ¥190,000
2020	**Joint Research of Research Center for Development of Far-Infrared Region, University of Fukui (FIR-UF)** Project Number: R02FIRDM016B Project Name: Investigation for novel ESR magnetic field standard at the very low–temperature using dilution refrigerator. Grant Amount: ¥190,000
2019	**Joint Research of Research Center for Development of FIR-UF** Project Number: R01FIRDM035A Project Name: Research for novel ESR magnetic field markers at cryogenic temperatures using a dilution refrigerator Grant Amount: ¥130,000
2019	**Public Interest Incorporated Association the Murata Science Foundation** The 34th (2019) Murata Science Foundation Overseas Dispatch Assistance Project No.: H31海自29, Dispatch Destination: EUROISMAR (21st ISMAR–15th EUROMAR Joint Conference, GDCh FGMR Discussion Meeting) | Berlin, Germany Grant Amount: ¥260,000
2019	**The Yukawa Memorial Foundation: Mochizuki Fund International Conference Dispatch** Destination: 60th Annual Rocky Mountain Conference on Magnetic Resonance | Denver, Colorado, USA Grant Amount: ¥250,000
2017–2018	**Development of the null detection method for spectacularly improvement of sensitivity of magnetic resonance measurements in terahertz region** (PI: Assoc. Prof. Susumu Okubo) Challenging Research (Exploratory) Grant-in-Aid for Scientific Research (KAKENHI)（Co-Investigator） Project/Area Number: 17K18760 Grant Amount: ¥6,240,000

Publications

I have authored 9 peer-reviewed publications including Journal Articles: 7 (including Review: 1), Proceedings: 2, Editorials: 0, Books / Chapters: 0 and Miscellaneous: 0.

Citations	h-index	i10-index
26	**4**	**0**

As of April 3, 2026 (Source: Google Scholar)

Peer-Reviewed Journal Articles:

† Denotes corresponding author, * Denotes co-first author

006	2019	Kunimoto, T., Fujita, Y., Kajiyama, H., Honma, T., Kawaguchi, H., Ishigaki, T., Ohmi, K., Ohshima, K., Miyazaki, D., **Saito, Y.**, Hara, S., Sakurai, T., Okubo, S. & Ohta, H. Development of Eu2+ and Mn2+ co-activated silicate phosphor for plant cultivation light source. *Jpn. J. Appl. Phys.* **58**, SFFD03 (2019). https://doi.org/10.7567/1347-4065/ab09cc	Cited by 4
005	2018	Okamoto, T., Ohmichi, E., **Saito, Y.**, Sakurai, T. & Ohta, H. Pressure Effect on Zero-Field Splitting Parameter of Hemin: Model Case of Hemoproteins under Pressure. *J. Phys. Chem. B* **122**, 6880–6887 (2018). https://doi.org/10.1021/acs.jpcb.8b03128	Cited by 6
004	2018	**Saito, Y.**†, Koseki, Y., Hatano, B., Sato, K., Okubo, S., Ohta, H. & Ogata, T. Reevaluation of analytical condition using highly singlet oxygen sensitive/selective reagent in aqueous-/non-aqueous solution. *Stud. Sci. Technol.* **7**, 61–68 (2018). https://doi.org/10.11425/sst.7.61	Cited by 1
003	2016	**Saito, Y.**, Nitto, Y., Hatano, B., Murakami, S., Ogata, T. & Kijima, T. Superoxide scavenging activity and colorimetric total polyphenol content analysis for traditional vegetables. Mater. Tech. 34, 159–165 (2016). (in Japanese).	Cited by —
002	2014	**Saito, Y.**†, Yanagisawa, K., Kimura, Y., Nitto, Y., Noda, H., Kijima, T. & Ogata, T. Pseudo flow-injection ESR technique combining spin-trapping and application to the evaluation of superoxide scavenging capacity of phenolic compound. *Stud. Sci. Technol.* **3**, 151–156 (2014). https://doi.org/10.11425/sst.3.151	Cited by —
001	2013	Fujita, N., **Saito, Y.***, Nitto, Y., Ito, T., Mizuguchi, H., Endo, M. & Ogata, T. Folin-Ciocalteu colorimetric analysis using a scanner for rapid determination of total polyphenol content in many test samples. *Stud. Sci. Technol.* **1**, 139–144 (2013). https://doi.org/10.11425/sst.1.139	Cited by —

Peer-Reviewed Proceedings:

002	2022	Koike, K., Uchida, T., Sakurai, K., Inaba, N., Kato, H., Itakura, M., Hara, S., **Saito, Y.**, Okubo, S. & Ohta, H. Infrared Laser Annealing of Nanocomposite Nd-Fe-B/Mo/FeCo Multilayered Magnet Films. *AIP Adv.* **12**, 035042 (2022). https://doi.org/10.1063/9.0000346	Cited by 3
001	2021	Koike, K., Ohashi, K., Suzuki, T., Okita, C., Inaba, N., Kato, H., Itakura, M., Hara, S., **Saito, Y.**, Okubo, S. & Ohta, H. Effect of Mo monoatomic interlayer on magnetic properties of in-plane anisotropic Nd-Fe-B/Mo/FeCo nanocomposite multilayered films. *AIP Adv.* **11**, 015134 (2021). https://doi.org/10.1063/9.0000181	Cited by 4

Review Articles:

001	2020	Sakurai, T., Ohta, H., Hara, S. & **Saito, Y.** High-Pressure THz ESR. *Appl. Magn. Reson.* **52**, 1–15 (2020). https://doi.org/10.1007/s00723-020-01299-7	Cited by 8

Editorials:

– N/A –

Books/Chapters:

– N/A –

Thesis

Dissertation	2015	Studies on Individual Reactive Oxygen Species Generation Systems and Their Application to the Evaluation of Reactive Oxygen Scavenging Activity (in Japanese)
Master Thesis	2012	Study on Singlet Oxygen Generation Systems in Non-aqueous Media and Its Application to Antioxidant Capacity Evaluation (in Japanese)
Bachelor Thesis	2008	Changes in Optical Properties of Retinoic Acid LB Films on ITO Films and Physical Properties of ITO Films Prepared by Sol-Gel/Dip-Coating Method (in Japanese)

Miscellaneous:

– N/A –

Communication

I have given 103 presentations: Intl. Conf. - 17 oral presentations (5 invited) and 11 poster presentations (1 invited), Dom. Conf. - 1 colloquia, 27 oral presentations, and 47 poster presentations.

indicates Invited

International Conference

– Oral Presentations

Invited Talks
(Lead Author)

– N/A –

(Co-author)

004	2021	Okubo, S., Tsuneishi, K., Takahashi, H., **Saito, Y.**, Hara, S., Sakurai, T., Ohmichi, E., Takahashi, K., Ohta, H., Nawa, K., Yajima, T., Okamoto, Y. & Hiroi, Z. “Multi-Frequency ESR Study of S=1/2 Antiferromagnetic Chain with Staggered Field System KCuMoO4(OH) by Force Detection ESR Method Using Single Microcrystal” Modern Development of Magnetic Resonance, Nov 1-5, 2021, Online.
003	2019	Okubo, S., Shimoshiro, S., **Saito, Y.**, Hara, S., Sakurai, T., Okamoto, T., Takahashi, H., Ohmichi, E., Ohta, H., Okuma, R. & Hiroi, Z. “High-field and High-frequency ESR Studies of Comprising Tetrahedral Clusters Arranged in the Cubic Lattice” Magnetic Resonance - Current State and Future Perspectives (EPR-75) at Kazan Federal University, Sep 23-27, 2019, Kazan, Russia.
002	2019	Ohta, H., Okubo, S., Ohmichi, E., Sakurai, T., Takahashi, H., Hara, S. & **Saito, Y.** “Multi-Extreme THz ESR - Recent Application and the Future -” Magnetic Resonance - Current State and Future Perspectives (EPR-75) at Kazan Federal University, Sep 23-27, 2019, Kazan, Russia.
001	2019	Ohta, H., Okubo, S., Ohmichi, E., Sakurai, T., Takahashi, H., Hara, S. & **Saito, Y.** “Recent Developments and Applications of Multi-extreme THz ESR” The 6th Awaji International Workshop on "Electron Spin Science & Technology: Biological and Materials Science Oriented Applications" (6th AWEST 2019), Jun 16-19, 2019, Awaji-shi, Japan.

Contributed Talks

(Lead author)

– N/A –

(Co-author)

002	2019	Ohta, H., Okubo, S., Ohmichi, E., Sakurai, T., Takahashi, H., Hara, S. & **Saito, Y.** “Multi-Extreme THz ESR: Developments on High-Pressure ESR and Mechanically Detected ESR” Rocky Mountain Conference on Magnetic Resonance, Jul 21-25, 2019, Colorado, USA.
001	2018	Okamoto, T., Ohmichi, E., **Saito, Y.**, Sakurai, T. & Ohta, H. “Pressure Effect on Zero-Field Splitting Parameter of Iron-Porphyrin Complexes Revealed by High-Frequency and High-Field Electron Paramagnetic Resonance” The Third Joint Conference of the Asia-Pacific EPR/ESR Society and The International EPR (ESR) Society (IES), Sep 23-27, 2018, Brisbane, Australia.

– Poster Presentations
(Lead author)

007	2024	**Saito, Y.**, Oba, E., Kimura, E., Ohyama, S., Nakagawa, T. & Arai, K. “NV-center based sensing in aqueous condition” International Workshop "Modern developments and applications of ESR, THz and high magnetic fields" (MDETH 2024), Sep 3-5, 2024, Kobe University, Hyogo, Japan.

006	2019	**Saito, Y.**, Sakurai, T., Hara, S., Okubo, S. & Ohta, H. “A High–Field ESR Study using Hexaaqua Complex Salt containing High–Spin Metal Ion: The Evaluation about a Potential of Tutton's Salt as a Pressure Standard for High–Pressure ESR Measurement” EUROISMAR 2019 (21st ISMAR–15th EUROMAR Joint Conference), Aug 25-30, 2019, Freie Universität Berlin, Berlin, Germany.
005	2019	**Saito, Y.**, Sakurai, T., Hara, S., Okubo, S. & Ohta, H. “A THz ESR Study under High-Pressure using Hexaaqua Complex Salt containing High–Spin Metal Ion” 60th Annual Rocky Mountain Conference on Magnetic Resonance, Jul 21-25, 2019, Denver, Colorado, USA.
004	2018	**Saito, Y.**, Koseki, Y., Hatano, B., Sato, K., Okubo, S., Ohta, H. & Ogata, T. “Antioxidant Capacity Evaluation of Carotenoid Compounds against Singlet Oxygen via ESR spectroscopy under in vitro Condition” The third joint conference of the Asia-Pacific EPR/ESR Society and The International EPR (ESR) Society (IES) Symposium 2018 (APES-IES 2018), Sep 23-27, 2018, the University of Queensland, Brisbane, Australia.
003	2014	**Saito, Y.**, Nitto, Y., Ohki, N., Sawara, T., Maruya, M. & Ogata, T. “Superoxide Scavenging Activity and Colorimetric Total Polyphenol Content Analysis for Traditional Vegetables” The Michinoku International Symposium on Analytical Science 2014, Aug 25-26, 2014, Sendai, Miyagi, Japan.
002	2013	**Saito, Y.**, Nakajima, K., Shiyama, T., Ogata, T. & Kijima, T. “ESR of Water Soluble Peroxyl Radical Generated from AIBN and Cyclodextrin” International Symposium for the 70th Anniversary of the Tohoku Branch of the Chemical Society of Japan 2013, Sep 28-30, 2013, Sendai, Japan.
001	2012	**Saito, Y.**, Koseki, Y., Watanabe, R., Fujita, N., Ito, T., Sato, K., Hatano, B. & Ogata, T. “Evaluation of Singlet Oxygen Scavenging and/or Quenching Effect of Antioxidants in Non-Aqueous Solvent” ISESS 2012 (The 2nd International Symposium on Electro Spin Science), Jul 23-25, 2012, Matsushima, Japan.

(Co-author)

004	2025	Okubo, S., Tsuneishi, K., Takahashi, H., **Saito, Y.**, Hara, S., Sakurai, T., Ohmichi, E., Ohta, H., Nawa, K., Yajima, T., Okamoto, Y. & Hiroi, Z. “Multi-frequency ESR Study of S=1/2 Antiferromagnetic Chain with Staggered Field System KCuMoO4(OH) single by Force Detection” 30th International Conference on Low Temperature Physics (LT30), Aug 7-13, 2025, Bilbao, Spain.
003	2019	Ohta, H., Okubo, S., Ohmichi, E., Sakurai, T., Takahashi, H., Hara, S. & **Saito, Y.** “Multi-Extreme THz ESR in Kobe” The 13th Japanese-Russian Workshop, Nov 10-13, 2019, Awaji-shi, Japan. (Invited)
002	2018	Okamoto, T., Ohmichi, E., **Saito, Y.**, Sakurai, T. & Ohta, H. “Pressure Effect on Zero-Field Splitting Parameter of Iron-Porphyrin Complexes Revealed by High-Frequency and High-Field Electron Paramagnetic Resonance” The Third Joint Conference of the Asia-Pacific EPR/ESR Society and The International EPR (ESR) Society (IES), Sep 23-27, 2018, Brisbane, Australia.
001	2012	Fujita, N., Yanagisawa, K., **Saito, Y.**, Ito, T. & Ogata, T. “Determination of antioxidant activities using electrogenerated superoxide” ISESS 2012 (The 2nd International Symposium on Electro Spin Science), Jul 23-25, 2012, Matsushima, Japan.

Domestic Conference

– Seminars / Colloquia

(Lead Author)

001	2018	**Saito, Y.** Analytical method of reactive oxygen and its application. **[Invited]** Paper presented at the 2nd SEST Young Researchers' Meeting (Kyoto Institute of Technology, 2018/07/20).

(Co-author)

– N/A –

– Oral Presentations

Invited Talks

(Lead Author)

– N/A –

(Co-author)

001		Okubo, S., Shimoshiro, S., **Saito, Y.**, Hara, S., Sakurai, T., Okamoto, T., Takahashi, H., Ohmichi, E., Ohta, H., Okuma, R. & Hiroi, Z. “High-Field ESR Measurements of the Fe-Tetrahedral Cluster Magnet Pharmacosiderite” Workshop on New Materials Science, Jul 8, 2019, Hokkaido University, Sapporo, Japan. (in Japanese)

Contributed Talks
(Lead author)

007	2024	**Saito, Y.**, Oba, E., Kimura, E., Ohyama, S., Nakagawa, T. & Arai, K. “The Trial of NV-based Quantum Sensor in Shore to Brackish water area ~The Blue Ocean Strategy for Coastal Resource Development~” 63rd Annual Meeting of the Society of Electron Spin Science and Technology (SEST2024), Nov 2-4, 2024, Fukui University, Fukui, Japan. (in Japanese)
006	2020	**Saito, Y.**, Ishikawa, Y., Sakurai, T., Okubo, S. & Ohta, H. “Low-temperature ESR measurements of ultra-stable perfluoroalkyl radicals” 59th Annual Meeting of the Society of Electron Spin Science and Technology (SEST2020), Nov 13-15, 2020, Online. (in Japanese)
005	2019	**Saito, Y.**, Okubo, S., Kitou, S., Taguchi, Y., Sawa, H. & Ohta, H. “X-Band study of Perovskite compound YTiO3: About the relationship of Ti-3d1 orbital state and rotation dependence” 58th Annual Meeting of the Society of Electron Spin Science and Technology (SEST2019), Nov 7-9, 2019, Kawasaki, Kanagawa, Japan. (in Japanese)

004	2018	**Saito, Y.**, Okamoto, T., Ohmichi, E., Sakurai, T., Hara, S., Okubo, S. & Ohta, H. “High-field ESR study of high-spin metal ion-type hexaaqua complex salts: Evaluation of Tutton's salt as a pressure standard for high-pressure ESR” 57th Annual Meeting of the Society of Electron Spin Science and Technology (SEST2018), Nov 1-3, 2018, Hokkaido University, Sapporo, Japan. (in Japanese)
003	2017	**Saito, Y.**, Okamoto, S., Oki, A., Sakurai, T., Hara, S., Ohmichi, E., Okubo, S. & Ohta, H. “Multi-frequency high-field ESR of hemin under pressure in the THz region” 56th Annual Meeting of the Society of Electron Spin Science and Technology (SEST2017), Nov 2-4, 2017, Tokyo Institute of Technology, Tokyo, Japan. (in Japanese)
002	2016	**Saito, Y.** “An attempt of reaction mechanism analysis by ESR for one-pot polymerization of aromatic olefins involving nitration” Institute for Molecular Science (IMS) Workshop - Advanced Electron Spin Measurements for the Origin of Biological and Material Functions, Dec 7-8, 2016, Okazaki, Aichi, Japan. (in Japanese)
001	2013	**Saito, Y.**, Nakajima, K., Ishiyama, T., Ito, T., Kijima, T. & Ogata, T. “ESR of water-soluble peroxyl radicals generated using AIBN/cyclodextrin” 52nd Annual Meeting of the Society of Electron Spin Science and Technology (SEST2013), Oct 24-26, 2013, Omiya, Saitama, Japan. (in Japanese)

(Co-author)

019	2022	Okubo, S., Tsuneishi, K., Takahashi, H., **Saito, Y.**, Hara, S., Sakurai, T., Ohmichi, E., Takahashi, K., Ohta, H., Nawa, K., Yajima, T., Okamoto, Y. & Hiroi, Z. “Single Crystal ESR Measurements of *S*=1/2 Antiferromagnetic Chain $KCuMoO_4(OH)$ IV” 77th JPS Annual Meeting, Mar 15-19, 2022, Online. (in Japanese)
018	2021	Okubo, S., Funakoshi, F., Takahashi, K., **Saito, Y.**, Hara, S., Sakurai, T., Takahashi, H., Ohmichi, E., Ohta, H., Kikuchi, H. & Fujii, Y. “High-Frequency ESR Measurements of Dimer Compound $CoSeO_3 \cdot 2H_2O$ IV” 76th JPS Annual Meeting, Mar 12-15, 2021, Online. (in Japanese)
017	2021	Okubo, S., Tsuneishi, K., Takahashi, H., **Saito, Y.**, Hara, S., Sakurai, T., Ohmichi, E., Takahashi, K., Ohta, H., Nawa, K., Yajima, T., Okamoto, Y. & Hiroi, Z. “Application of Terahertz Electron Spin Resonance to Low-Dimensional Magnets” 18th Annual Meeting of the Association of Infrared Radiation Applications, Jan 29, 2021, Online. (in Japanese)
016	2020	Okubo, S., Suzuki, T., **Saito, Y.**, Hara, S., Sakurai, T., Takahashi, H., Ohmichi, E., Ohta, H., Matsumoto, M., Kurita, N. & Tanaka, H. “Study of Pressure-Induced Magnetic Phase of Spin-Gap System $CsFeCl_3$ by ESR Measurements under Pressure” 59th Annual Meeting of the Society of Electron Spin Science and Technology (SEST2020), Nov 13-15, 2020, Online. (in Japanese)
015	2020	Okubo, S., Suzuki, T., **Saito, Y.**, Hara, S., Sakurai, T., Takahashi, H., Ohmichi, E., Ohta, H., Matsumoto, M., Kurita, N. & Tanaka, H. “Study of Triangular Lattice Antiferromagnet $CsFeCl_3$ by Terahertz ESR under Pressure” 29th Annual Meeting of the Japan Society of Infrared Science and Technology, Oct 29-30, 2020, Online (in Japanese)

014	2020	Okubo, S., Suzuki, T., **Saito, Y.**, Hara, S., Sakurai, T., Ohta, H., Matsumoto, M., Kurita, N. & Tanaka, H. “ESR Study of Pressure-Induced Magnetic Phase of $CsFeCl_3$ III” JPS 2020 Autumn Meeting, Sep 8-11, 2020, Online. (in Japanese)
013	2020	Funakoshi, F., Takahashi, K., **Saito, Y.**, Hara, S., Sakurai, T., Takahashi, H., Okubo, S., Ohta, H., Kikuchi, H. & Fujii, Y. “High-Frequency ESR Measurements of Dimer Compound $CoSeO_3{\cdot}2H_2O$ III” JPS 2020 Autumn Meeting, Sep 8-11, 2020, Online(in Japanese)
012	2020	Tsuneishi, K., Takahashi, H., **Saito, Y.**, Hara, S., Sakurai, T., Okubo, S., Ohta, H., Ohmichi, E., Takahashi, K., Nawa, K., Yajima, T., Okamoto, Y. & Hiroi, Z. “Single Crystal ESR Measurements of *S*=1/2 Antiferromagnetic Chain $KCuMoO_4(OH)$ III” JPS 2020 Autumn Meeting, Sep 8-11, 2020, Online. (in Japanese)
011	2019	Okubo, S., Shimoshiro, S., **Saito, Y.**, Hara, S., Sakurai, T., Okamoto, T., Takahashi, H., Ohmichi, E., Ohta, H., Okuma, R. & Hiroi, Z. “Study of Fe-Cluster Antiferromagnets by Terahertz ESR” 28th Annual Meeting of the Japan Society of Infrared Science and Technology, Nov 14, 2019, Fukui University, Fukui, Japan. (in Japanese)
010	2019	Tsuneishi, K., Takahashi, H., **Saito, Y.**, Hara, S., Sakurai, T., Ohmichi, E., Takahashi, K., Okubo, S., Ohta, H., Nawa, K., Yajima, T., Okamoto, Y. & Hiroi, Z. “Single Crystal ESR Measurements of *S*=1/2 Antiferromagnetic Chain $KCuMoO_4(OH)$” JPS 2019 Autumn Meeting, Sep 10-13, 2019, Gifu University, Gifu, Japan.(in Japanese)
009	2019	Funakoshi, F., **Saito, Y.**, Hara, S., Sakurai, T., Okubo, S., Ohta, H., Fujii, Y., Kato, K. & Kikuchi, H. “High-Frequency ESR Measurements of Dimer Compound $CoSeO_3{\cdot}2H_2O$” JPS 2019 Autumn Meeting, Sep 10-13, 2019, Gifu University, Gifu, Japan.(in Japanese)
008	2019	Oshima, K., **Saito, Y.**, Hara, S., Sakurai, T., Okubo, S., Ohta, H., Yamamoto, H. & Azuma, M. “High-Frequency ESR Measurements of Perovskite Compound $PbMnO_3$ II” JPS 2019 Autumn Meeting, Sep 10-13, 2019, Gifu University, Gifu, Japan. (in Japanese)
007	2019	Okubo, S., Shimoshiro, S., **Saito, Y.**, Hara, S., Sakurai, T., Okamoto, T., Takahashi, H., Ohmichi, E., Ohta, H., Okuma, R. & Hiroi, Z. “High-Field ESR Study of Antiferromagnets with Geometrical Spin Structures IV” JPS 2019 Autumn Meeting, Sep 10-13, 2019, Gifu University, Gifu, Japan. (in Japanese)
006	2019	Suzuki, T., Okuto, R., **Saito, Y.**, Hara, S., Sakurai, T., Takahashi, H., Okubo, S., Ohta, H., Matsumoto, M., Kurita, N. & Tanaka, H. “ESR Study of Pressure-Induced Magnetic Phase of $CsFeCl_3$” JPS 2019 Autumn Meeting, Sep 10-13, 2019, Gifu University, Gifu, Japan.(in Japanese)
005	2018	Okubo, S., Shimoshiro, S., **Saito, Y.**, Hara, S., Sakurai, T., Okamoto, T., Takahashi, H., Ohmichi, E., Ohta, H., Okuma, R. & Hiroi, Z. “High-Field ESR Study of Antiferromagnets with Geometrical Spin Structures III” 74th JPS Annual Meeting, Mar 14-17, 2019, Kyushu University, Fukuoka, Japan.(in Japanese)

004	2018	Okubo, S., Miyazaki, D., Okamoto, S., Takahashi, H., Ohmichi, E., **Saito, Y.**, Hara, S., Sakurai, T., Ohta, H., Irisawa, A. & Fujita, W. "ESR Measurements of *S*=1/2 Distorted Diamond Chain Compound ($Cu_3(OH)_2(CH_3CO_2)_2(H_2O)_xX_y$) II" 57th Annual Meeting of the Society of Electron Spin Science and Technology (SEST2018), Nov 1-3, 2018, Hokkaido University, Sapporo, Japan. (in Japanese)
003	2018	Ohta, H., Okubo, S., Ohmichi, E., Sakurai, T., Takahashi, H., Hara, S. & **Saito, Y.** "Developments and Perspectives of Multi-Extreme THz ESR" 57th Annual Meeting of the Society of Electron Spin Science and Technology (SEST2018), Nov 1-3, 2018, Hokkaido University, Sapporo, Japan.(in Japanese)
002	2018	Shimoshiro, S., Okubo, S., **Saito, Y.**, Hara, S., Sakurai, T., Okamoto, T., Takahashi, H., Ohmichi, E., Ohta, H., Okuma, R. & Hiroi, Z. "High-Field ESR Study of Antiferromagnets with Geometrical Spin Structures II" JPS 2018 Autumn Meeting, Sep 9-12, 2018, Doshisha University, Kyoto, Japan. (in Japanese)
001	2018	Okubo, S., Miyazaki, D., Benzid, K., **Saito, Y.**, Hara, S., Sakurai, T., Ohta, H., Irisawa, A. & Fujita, W. "Submillimeter Wave ESR Measurements of *S*=1/2 Distorted Diamond Chain Compound ($Cu_3(OH)_2(CH_3CO_2)_2(H_2O)_xX_y$) III" JPS 2018 Autumn Meeting, Sep 9-12, 2018, Doshisha University, Kyoto, Japan. (in Japanese)

– Poster Presentations

(Lead author)

008	2019	**Saito, Y.**, Koseki, Y., Hatano, B., Sato, K., Okubo, S., Ohta, H. & Ogata, T. "Antioxidant Capacity Evaluation Carotenoids against Singlet Oxygen via ESR Spectroscopy under *in vitro* condition" The 99th CSJ Annual Meeting, Mar 16-19, 2019, Konan University, Kobe, Hyogo, Japan. (in Japanese)
007	2018	**Saito, Y.**, Miyazaki, D., Benzid, K., Okubo, S., Hara, S., Sakurai, T., Ohta, H., Matsuoka, E., Sugawara, J. & Fujita, W. "High-Field ESR Study of $Cu_3(OH)_2(OAc)_2(H_2O)_x(2,6\text{-NpSA})_y$ (2,6-NpSA; 2,6-Naphthalenesulfonic Acid) as Copper Hydroxy Sulfonates with *S*=1/2 Distorted Diamond Chain Networks" The 5th Meeting on High Magnetic Field Science in Western Japan, Sep 6, 2018, Kobe University, Kobe, Japan. (in Japanese)
006	2016	**Saito, Y.**, Ogata, T., Murakami, S., Hatano, B. & Kijima, T. "Reaction mechanism analysis of one-pot polymerization with nitration using styrene by ESR spectroscopy" 55th Annual Meeting of the Society of Electron Spin Science and Technology (SEST2016), Nov 10-12, 2016, Osaka City University, Osaka, Japan. (in Japanese)
005	2016	**Saito, Y.**, Murakami, S., Hatano, B. & Kijima, T. "Reaction mechanism analysis of one-pot polymerization by ESR spectroscopy" Joint Meeting of the Tohoku Area Chemistry Societies 2016, Sep 10-11, 2016, Iwaki Meisei University, Iwaki, Fukushima, Japan. (in Japanese)

004	2012	**Saito, Y.**, Koseki, Y., Fujita, N., Sato, K., Hatano, B. & Ogata, T. “Evaluation of antioxidant capacity of carotenoids using singlet oxygen generated by photosensitizers and endoperoxides” 51st Annual Meeting of the Society of Electron Spin Science and Technology (SEST2012), Nov 1-3, 2012, Hokkaido University, Sapporo, Japan. (in Japanese)
003	2012	**Saito, Y.**, Koseki, Y., Watanabe, R., Fujita, N., Ito, T., Sato, K., Hatano, B. & Ogata, T. “Investigation of singlet oxygen generation system using naphthalene derivative endoperoxides in non-aqueous solvents and its application to antioxidant capacity evaluation” Joint Meeting of the Tohoku Area Chemistry Societies 2012, Sep 15-16, 2012, Akita University, Akita, Japan. (in Japanese)
002	2012	**Saito, Y.**, Koseki, Y., Hatano, B. & Ogata, T. “Investigation of singlet oxygen generation system in non-aqueous solvents using photosensitization and evaluation of antioxidant capacity” Michinoku International Symposium on Analytical Science 2012, Jul 21, 2012, Yamagata University, Yonezawa, Yamagata, Japan. (in Japanese)
001	2011	**Saito, Y.**, Koseki, Y., Hatano, B., Ito, T. & Ogata, T. “Study on the evaluation method of singlet oxygen scavenging activity in non-aqueous solvents” Michinoku International Symposium on Analytical Science 2011, Jul 23, 2011, Yamagata University, Yonezawa, Yamagata, Japan. (in Japanese)

(Co-author)

039	2020	Funakoshi, F., **Saito, Y.**, Sakurai, T., Takahashi, H., Ohmichi, E., Okubo, S., Ohta, H., Kikuchi, H., Fujii, Y. & Hara, S. "Terahertz-wave ESR measurements of the spin dimer compound $CoSeO_3 \cdot 2H_2O$." The 29th Annual Meeting of the Japan Society of Infrared Science and Technology, Oct 29–30, 2020, Online. (in Japanese)
038	2020	Tsuneishi, K., Takahashi, H., **Saito, Y.**, Hara, S., Sakurai, T., Ohmichi, E., Takahashi, K., Okubo, S., Ohta, H., Nawa, K., Yajima, T., Okamoto, Y. & Hiroi, Z. "Terahertz force-detected ESR measurements of the S=1/2 antiferromagnetic chain $KCuMoO_4(OH)$ using a micro-single crystal." The 29th Annual Meeting of the Japan Society of Infrared Science and Technology, Oct 29–30, 2020, Online. (in Japanese)
037	2020	Tsuneishi, K., Takahashi, H., **Saito, Y.**, Hara, S., Sakurai, T., Ohmichi, E., Takahashi, K., Okubo, S., Ohta, H., Nawa, K., Yajima, T., Okamoto, Y. & Hiroi, Z. "Membrane force-detected ESR measurements of the S=1/2 antiferromagnet $KCuMoO_4(OH)$ using a micro-single crystal." The 59th Annual Meeting of the Society of Electron Spin Science and Technology (SEST2020), Nov 13–15, 2020, Online, PS18. (in Japanese)
036	2020	Funakoshi, F., **Saito, Y.**, Sakurai, T., Takahashi, H., Ohmichi, E., Okubo, S., Ohta, H., Kikuchi, H., Fujii, Y. & Hara, S. "High-frequency ESR measurements of the spin dimer compound $CoSeO_3 \cdot 2H_2O$." The 59th Annual Meeting of the Society of Electron Spin Science and Technology (SEST2020), Nov 13–15, 2020, Online, PS9. (in Japanese)
035	2019	Funakoshi, F., **Saito, Y.**, Hara, S., Sakurai, T., Okubo, S., Ohta, H., Takahashi, K., Kikuchi, H. & Fujii, Y. "High-frequency ESR measurements of the spin dimer compound $CoSeO_3 \cdot 2H_2O$." Wakate Frontier Kenkyukai 2019, Dec 19, 2019, Kobe, Japan. (in Japanese)

034	2019	Ohshima, K., Sakurai, T., Hara, S., **Saito, Y.**, Okubo, S., Ohta, H., Yamamoto, I. & Azuma, M. "ESR measurements of lead-type perovskite compounds." Wakate Frontier Kenkyukai 2019, Dec 19, 2019, Kobe, Japan. (in Japanese)
033	2019	Suzuki, T., **Saito, Y.**, Hara, S., Sakurai, T., Okubo, S., Ohta, H., Kurita, N., Tanaka, H. & Matsumoto, M. "High-field ESR study of the pressure-induced magnetic phase in $CsFeCl_3$." The 15th High Magnetic Field Forum, Dec 3–5, 2019, Toyonaka, Japan. (in Japanese)
032	2019	Ohshima, K., Sakurai, T., Hara, S., **Saito, Y.**, Okubo, S., Ohta, H., Yamamoto, I. & Azuma, M. "High-field measurements of perovskite compounds $PbMO_3$ (M=Mn, Cr, V)." The 15th High Magnetic Field Forum, Dec 3–5, 2019, Toyonaka, Japan. (in Japanese)
031	2019	Tsuneishi, K., Takahashi, H., **Saito, Y.**, Hara, S., Sakurai, T., Ohmichi, E., Takahashi, K., Okubo, S., Ohta, H., Nawa, K., Yajima, T., Okamoto, Y. & Hiroi, Z. "Observation of magnetic anisotropy in the S=1/2 antiferromagnetic chain $KCuMoO_4(OH)$ with alternating D-vectors using high-field ESR on a micro-single crystal." The 15th High Magnetic Field Forum, Dec 3–5, 2019, Toyonaka, Japan. (in Japanese)
030	2019	Funakoshi, F., **Saito, Y.**, Hara, S., Sakurai, T., Okubo, S., Ohta, H., Kikuchi, H. & Fujii, Y. "High-field ESR measurements of the spin dimer compound $CoSeO_3{\cdot}2H_2O$." The 15th High Magnetic Field Forum, Dec 3–5, 2019, Toyonaka, Japan. (in Japanese)
029	2019	Okubo, S., Shimoshiro, S., **Saito, Y.**, Hara, S., Sakurai, T., Okamoto, S., Takahashi, H., Ohmichi, E., Ohta, H., Okuma, R. & Hiroi, Z. "High-field ESR measurements of Fe-tetrahedral cluster antiferromagnets." The 15th High Magnetic Field Forum, Dec 3–5, 2019, Toyonaka, Japan. (in Japanese)
028	2019	Okubo, S., Miyazaki, D., Okamoto, S., Takahashi, H., Ohmichi, E., **Saito, Y.**, Hara, S., Sakurai, T., Ohta, H., Irisawa, A. & Fujita, W. "ESR study of the ground state of S=1/2 distorted diamond chain compounds." The 13th Quantum Spin System Workshop, Jan 9–10, 2019, Tokai, Japan. (in Japanese)
027	2018	Tsuneishi, K., **Saito, Y.**, Hara, S., Sakurai, T., Okubo, S. & Ohta, H. "ESR measurements of $KCuMoO_4(OH)$." Wakate Frontier Kenkyukai 2018, Dec 21, 2018, Kobe, Japan. (in Japanese)
026	2018	Ohshima, K., **Saito, Y.**, Hara, S., Okubo, S., Ohta, H., Yamamoto, I. & Azuma, M. "ESR measurements of the perovskite-type compound $PbMnO_3$." Wakate Frontier Kenkyukai 2018, Dec 21, 2018, Kobe, Japan. (in Japanese)
025	2018	Shimoshiro, S., Okubo, S., **Saito, Y.**, Hara, S., Sakurai, T., Okamoto, S., Takahashi, H., Ohmichi, E., Ohta, H., Okuma, R. & Hiroi, Z. "High-field ESR study of antiferromagnets with geometric spin structures." Wakate Frontier Kenkyukai 2018, Dec 21, 2018, Kobe, Japan. (in Japanese)
024	2018	Okubo, S., Suzuki, T., Bruand, P., Lassagne, B., **Saito, Y.**, Hara, S., Sakurai, T. & Ohta, H. "Terahertz electron spin resonance measurements by the null detection method." Workshop on Present Status and Future Prospects of Material Science in High Magnetic Field Collaboratory, Nov 26–28, 2018, Sendai, Japan. (in Japanese)

023	2018	Suzuki, T., **Saito, Y.**, Hara, S., Sakurai, T., Okubo, S., Ohta, H., Akagi, Y., Narumi, Y., Hagiwara, M. & Yoshida, H. "Magnetism under pressure in the S=1/2 kagome lattice antiferromagnet Ca-kapellasite." Physical Society of Japan 2018 Autumn Meeting, Sep 9–12, 2018, Kyotanabe, Japan, 9aPS-65. (in Japanese)
022	2018	Ohshima, K., **Saito, Y.**, Hara, S., Sakurai, T., Okubo, S., Ohta, H., Yamamoto, I. & Azuma, M. "High-frequency ESR measurements of the perovskite-type compound $PbMnO_3$." Physical Society of Japan 2018 Autumn Meeting, Sep 9–12, 2018, Kyotanabe, Japan, 9pPSA-65. (in Japanese)
021	2018	Ohta, H., Okubo, S., Takahashi, H., Ohmichi, E., **Saito, Y.**, Hara, S. & Sakurai, T. "Multi-extreme THz ESR at Kobe University." The 5th Meeting on High Magnetic Field Science in Western Japan, Sep 6, 2018, Kobe, Japan. (in Japanese)
020	2018	Bruand, P., Lassagne, B., Suzuki, T., Okubo, S., **Saito, Y.**, Hara, S., Sakurai, T. & Ohta, H. "Development of High-field ESR system using the null detection." The 5th Meeting on High Magnetic Field Science in Western Japan, Sep 6, 2018, Kobe, Japan. (in Japanese)
019	2018	Ohshima, K., **Saito, Y.**, Hara, S., Sakurai, T., Okubo, S., Ohta, H., Yamamoto, I. & Azuma, M. "High magnetic field ESR measurement of perovskite type compound $PbMnO_3$." The 5th Meeting on High Magnetic Field Science in Western Japan, Sep 6, 2018, Kobe, Japan. (in Japanese)
018	2018	Suzuki, T., **Saito, Y.**, Hara, S., Sakurai, T., Okubo, S., Ohta, H., Akagi, Y., Narumi, Y., Hagiwara, M. & Yoshida, H. "Susceptibility measurements under pressure in the S=1/2 kagome lattice antiferromagnet Ca-kapellasite." The 5th Meeting on High Magnetic Field Science in Western Japan, Sep 6, 2018, Kobe, Japan. (in Japanese)
017	2018	Shimoshiro, S., Okubo, S., **Saito, Y.**, Hara, S., Sakurai, T., Okamoto, S., Takahashi, H., Ohmichi, E., Ohta, H., Okuma, R. & Hiroi, Z. "High-field ESR measurements of antiferromagnets with geometric spin structures." The 5th Meeting on High Magnetic Field Science in Western Japan, Sep 6, 2018, Kobe, Japan. (in Japanese)
016	2018	Hara, S., Fujimoto, M., Okubo, S., **Saito, Y.**, Sakurai, T., Ohta, H., Otsuka, D. & Sato, H. "High-field ESR measurements of the spin frustrated magnet $K_2Mn_3(OH)_2(VO_4)_2$." The 5th Meeting on High Magnetic Field Science in Western Japan, Sep 6, 2018, Kobe, Japan. (in Japanese)
015	2018	Ohta, H., Okubo, S., Takahashi, H., Ohmichi, E., **Saito, Y.**, Hara, S. & Sakurai, T. "Multi-extreme THz ESR in Kobe Univ. – the high-sensitive membrane ESR and the high pressure ESR –." The 5th Meeting on High Magnetic Field Science in Western Japan, Sep 6, 2018, Kobe, Japan, P-26. (in Japanese)
014	2018	Hara, S., Fujimoto, M., Okubo, S., **Saito, Y.**, Sakurai, T., Ohta, H., Otsuka, D. & Sato, H. "High field ESR measurements of spin frustrated magnet $K_2Mn_3(OH)_2(VO_4)_2$." The 5th Meeting on High Magnetic Field Science in Western Japan, Sep 6, 2018, Kobe, Japan, P-25. (in Japanese)

013	2018	Bruand, P., Lassagne, B., Suzuki, T., Okubo, S., **Saito, Y.**, Hara, S., Sakurai, T. & Ohta, H. "Development of High-field ESR system using the null detection." The 5th Meeting on High Magnetic Field Science in Western Japan, Sep 6, 2018, Kobe, Japan, P-22. (in Japanese)
012	2018	Ohshima, K., **Saito, Y.**, Hara, S., Sakurai, T., Okubo, S., Ohta, H., Yamamoto, I. & Azuma, M. "High magnetic field ESR measurement of perovskite type compound $PbMnO_3$." The 5th Meeting on High Magnetic Field Science in Western Japan, Sep 6, 2018, Kobe, Japan, P-21. (in Japanese)
011	2018	Suzuki, T., **Saito, Y.**, Hara, S., Sakurai, T., Okubo, S., Ohta, H., Akagi, Y., Narumi, Y., Hagiwara, M. & Yoshida, H. "ESR measurement of S=1/2 kagome lattice antiferromagnet Ca-kapellasite." The 5th Meeting on High Magnetic Field Science in Western Japan, Sep 6, 2018, Kobe, Japan, P-20. (in Japanese)
010	2018	Shimoshiro, S., Okubo, S., **Saito, Y.**, Hara, S., Sakurai, T., Okamoto, S., Takahashi, H., Ohmichi, E., Ohta, H., Okuma, R. & Hiroi, Z. "High field ESR measurement of geometrical spin structured antiferromagnet." The 5th Meeting on High Magnetic Field Science in Western Japan, Sep 6, 2018, Kobe, Japan, P-15. (in Japanese)
009	2018	Shimoshiro, S., Okubo, S., **Saito, Y.**, Hara, S., Sakurai, T., Ohta, H., Okuma, R. & Hiroi, Z. "High-field ESR study of antiferromagnets with geometric spin structures." Physical Society of Japan 73rd Annual Meeting, Mar 22–25, 2018, Noda, Japan, 22pPSA-36. (in Japanese)
008	2017	Shimoshiro, S., Takahashi, A., Yoshida, S., Benzid, K., **Saito, Y.**, Hara, S., Sakurai, T., Okubo, S., Ohta, H. & Sato, H. "ESR measurements of Co square-lattice antiferromagnets." Wakate Frontier Kenkyukai, Dec 21, 2017, Kobe, Japan, C03. (in Japanese)
007	2017	Shimoshiro, S., Takahashi, A., Yoshida, S., Benzid, K., **Saito, Y.**, Hara, S., Sakurai, T., Okubo, S., Ohta, H. & Sato, H. "High-field ESR measurements of Co square-lattice antiferromagnets." The 4th Meeting on High Magnetic Field Science in Western Japan, Sep 25, 2017, Osaka, Japan, P-12. (in Japanese)
006	2017	Shimoshiro, S., Takahashi, A., Yoshida, S., Benzid, K., **Saito, Y.**, Hara, S., Sakurai, T., Okubo, S., Ohta, H. & Sato, H. "ESR measurements of Co square-lattice." Physical Society of Japan 2017 Autumn Meeting, Sep 21–24, 2017, Morioka, Japan, 21aPS-11. (in Japanese)
005	2017	Nitto, Y., Yanagisawa, K., Kimura, Y., **Saito, Y.**, Itoh, T. & Ogata, T. "Folin-Ciocalteu analysis and superoxide scavenging activity analysis of phenolic compounds." The 52nd Annual Meeting of the Society of Electron Spin Science and Technology (SEST2013), Oct 24–26, 2013, Omiya, Japan, pp. 144–145. (in Japanese)
004	2013	Fujita, N., Nitto, Y., **Saito, Y.**, Watanabe, R., Itoh, T. & Ogata, T. "Study on multi-sample simultaneous colorimetric analysis of polyphenol-containing samples using a scanner." 2012 Tohoku Conference of Chemistry-related Societies, Sep 15–16, 2012, Akita, Japan, 2P019. (in Japanese)

003	2012	Koseki, H., **Saito, Y.**, Itoh, T., Sato, K., Hatano, T. & Ogata, T. "Study on singlet oxygen generation systems and their application to antioxidant capacity evaluation." 2012 Tohoku Conference of Chemistry-related Societies, Sep 15–16, 2012, Akita, Japan, 1P022. (in Japanese)
002	2012	Nitto, Y., Oki, N., **Saito, Y.**, Fujita, N. & Ogata, T. "Study on multi-sample simultaneous colorimetric analysis using a scanner." Michinoku Symposium on Analytical Sciences 2012, Jul 21, 2012, Yonezawa, Japan, P41. (in Japanese)
001	2012	Koseki, H., **Saito, Y.**, Sato, K., Hatano, T. & Ogata, T. "Study on singlet oxygen generation systems in non-aqueous solvents using naphthalene derivative endoperoxides and antioxidant capacity evaluation." Michinoku Symposium on Analytical Sciences 2012, Jul 21, 2012, Yonezawa, Japan, P40. (in Japanese)

Teaching Experience

I have taught 6 modules: 6 as Instructor, Staff, Research Assistant, or Teaching Assistant.

2024 – 2025 **Safety Training on Laser and High Power Sources** | *Tokyo Institute of Technology / International Christian University*
Role: On-site Instructor
Target: 5+ Bachelor and Master's students per session. **Content**: Provided specialized hands-on safety training for high-power laser systems and electrical sources.

2017 - 2021 **University-wide Safety Training on Cryogens (LN_2 and LHe)** | *Kobe University*
Role: Operational Staff
Target:450+ users annually, including undergraduate/graduate students, faculty (Professors, Associate Professors, Assistant Professors, Lecturers), technicians, postdoctoral fellows, and senior scientists.
Key Achievement: Maintained a zero-incident record across all training sessions. Managed two critical crisis management cases effectively.
Responsibilities: Distribution of legal compliance materials, attendance management, and regulatory documentation.

2017 - 2021 **University-wide Safety Training on Radiative Equipment and Materials** | Kobe University
Role:: Support Staff
Target: 600+ users annually, spanning all academic and technical ranks from students to senior faculty.
Responsibilities: Distribution of radiation safety protocols and materials, and verification of legal attendance requirements.

2017 - 2021 **Laboratory Safety Training on Microwaves, Lasers, High-Voltage Capacitors, and Cryogenics** | *Kobe University*
Role: On-site Instructor
Target: 10+ Bachelor and Master's students.
Content: Instructed on the handling and transport of high-pressure vessels, microwave safety, and high-voltage capacitor operations.

2012 – 2015	**Inorganic Chemistry I/II (Faculty-wide Courses)** | *Yamagata University* **Role**: Research Assistant and Teaching Assistant **Target**: 6 undergraduate classes over 3 years (120+ students per class). **Responsibilities**: Assisted in lecture delivery, managed student inquiries, proctored examinations, and conducted grading and attendance tracking.
2010 - 2013	**Elementary School Science Lessons** | *Yonezawa City Board of Education* **Role**: Project Science Instructional and Lab Assistant (Science Supporter) **Target**: 5th and 6th-grade students (annually 6 classes, 30 children per class). **Responsibilities**: Fostered scientific curiosity and understanding in children.

Mentoring & Supervision

Extensive mentorship experience, having supervised over 230 students (total) in laboratory safety protocols and advanced research methodologies.

2024 – 2025	**Specially Appointed Postdoctoral Researcher** | *International Christian University* **Impact**: Mentored 5+ undergraduate students annually, all of whom successfully obtained their Bachelor's degrees. **Scope**: Guided students in experimental procedures, manuscript preparation, and oral/poster presentations at conferences. Conducted safety orientations for hazardous materials, including cryogens, X-rays, and high-voltage systems.
2023 - 2024	**Specially Appointed Postdoctoral Researcher** | *Tokyo Institute of Technology* **Impact**: Mentored 10 laboratory students annually (6 graduate, 4 undergraduate), leading to 5 Bachelor's and 4 Master's degrees. **Scope**: Provided technical guidance on experimental workflows and academic writing. Ensured rigorous safety compliance for cryogenics and radiation equipment.
2023 - Present	**Visiting Researcher** | *Kanagawa University* **Impact**:Mentors 15 laboratory students annually (10 graduate, 5 undergraduate), resulting in 10 Bachelor's and 5 Master's degrees per cycle. **Scope**:Direct supervision of research projects, conference presentation training, and comprehensive safety management of laboratory equipment.
2021 - 2023	**Postdoctoral Fellow (NEDO Project)** | *Sophia University* **Impact**: Mentored 35 laboratory students annually (20 graduate, 15 undergraduate), supporting 20 Bachelor's and 15 Master's degree completions. **Scope**: Managed large-scale student cohorts within a national research project (NEDO), focusing on high-level experimental training and manuscript submission.
2017 - 2021	**Postdoctoral Researcher** | *Kobe University* **Impact**:Mentored 15–20 laboratory students annually (10–13 graduate, 5–7 undergraduate). **Scope**:Intensive mentorship in physical chemistry and spectroscopy, ensuring all students met graduation requirements and presentation standards.

2015 - 2017	**Postdoctoral Researcher** | *Yamagata University* **Impact**:Supervised 20+ students annually, achieving a 100% graduation rate through intensive training in organic and physical chemistry. **Scope**:Foundational mentorship in experimental design and safety protocols for diverse hazardous materials.

Public Outreach & Social Contribution

2012 - 2017	**Science Workshops for High School Students (SSH & Partner Schools)** | *Yamagata University* **Role**: Instructor **Frequency**: Biweekly during the semester (average 20–22 sessions per year). **Content**: Delivered outreach lectures and hands-on workshops for Super Science High School (SSH) partners and partner schools.
2012 - 2017	**On-site Science Workshops for Junior High School Students** | *Yamagata University* **Role**: Instructor **Frequency**: 10+ sessions annually.
2008 - 2017	**Mobile Kids Chemi-Lab (Mobile Chemistry Workshop)** | *Yamagata University and Yonezawa City Board of Education* **Role**: Instructor **Frequency**: 10–20 sessions annually (including 8+ regular and on-demand sessions).
2011	**Slime Meister Course (Yamagata "Kagaku no Hana Saku" Project)** | *Yamagata University and Yonezawa City Board of Education* **Role**: Instructor **Frequency**: 10 sessions

Professional Qualifications Memberships Social Activities

2018 – Present:	International ESR(EPR) Society (IES)
2018 – Present:	The International Society of Magnetic Resonance (ISMAR)
2017 – Present:	The Chemical Society of Japan
2010 – Present:	The Society of Electron Spin Science and Technology (SEST)
2010 – 2017:	The Japan Society for Analytical Science

Technical Skills

Languages

Japanese:	Native speaker
English:	Intermediate to Proficien (Experienced in international conferences, collaborative research with international scholars, and professional/daily communication)

Technical Proficiency

Operating Systems:	Mac (Power User), Windows (Advanced), Linux (Basic)
Apps, Dev. & Prog.:	<Advance> MATLAB, Igor Pro, Python, Microsoft Office Suite, Adobe Illustrator, Adobe Photoshop, Markdown, Jupyter Notebook <Proficient> Gaussian, WIEN2k, ORCA (Quantum Chemical Calculations), R, Mathematica, SciLab, Julia

Analytical Instrumentation

Expertise:	Electron Spin Resonance (ESR / EPR)
Hands-on Experience:	NMR, HPLC, GC-MS, FT-IR, UV-Vis, XRD, ICP-MS/AES, ESI-MS, TEM, SEM

Research Expertise

Experimental Skills:	High-pressure and high-magnetic field measurements, cryogenic experimental setup using liquid *He* and N_2.
Cryogen Handling:	Safe handling and refilling of liquid helium/nitrogen, management of cryogenic systems and superconducting magnets.

Certifications & Licenses

– **Doctor of Philosophy in Engineering** / Yamagata University (2015)

– **Master of Engineering** / Yamagata University (2012)

– **Bachelor of Engineering** / Akita University (2008)

JABEE Accredited Program Completed, Materials Science and Engineering

– **High School Teacher's Specialized Certificate (Industry)** (2012)

– **High School Teacher's First-class Certificate (Industry)** (2008)

– **Medium-sized Motor Vehicle License (8t limited)** (2004)